Recall

Measures of Central Tendency

It is often convenient to have one number that represent the whole data. Such a number is called a Measures of Central Tendency.

The Measures of Central Tendency usually will be near to the middle value of the data. For a given data there exist several types of measures of central tendencies.

The most common among them are

- Arithmetic Mean
- Median
- Mode

Thinking Corner

1. Does the mean, median and mode are same for a given data?
2. What is the difference between the arithmetic mean and average?

Note

- Data : The numerical representation of facts is called data.
- Observation : Each entry in the data is called an observation.
- Variable : The quantities which are being considered in a survey are called variables. Variables are generally denoted by x_i, $i = 1,2,3,\ldots,n$.
- Frequencies : The number of times, a variable occurs in a given data is called the frequency of that variable. Frequencies are generally denoted as f_i, $i = 1,2,3,\ldots,n$.

In this class we have to recall the Arithmetic Mean.

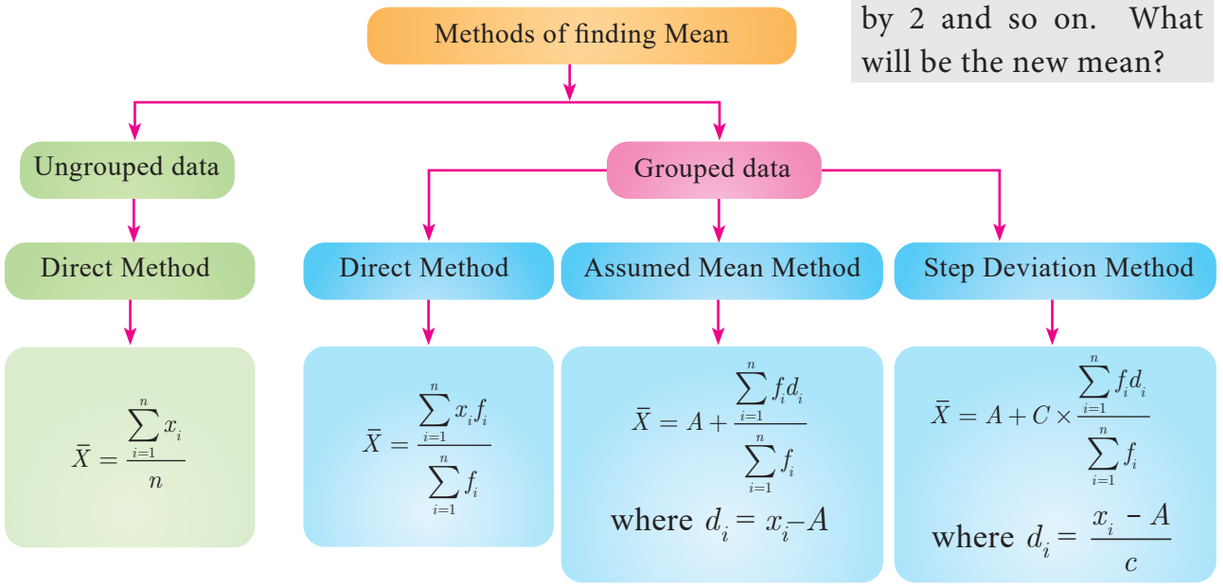
Arithmetic Mean

The Arithmetic Mean or Mean of the given values is sum of all the observations divided by the total number of observations. It is denoted by $\bar{x}$ (pronounced as x bar)

$$\bar{x} = \frac{\text{Sum of all the observations}}{\text{Number of observations}}$$

Thinking Corner

The mean of n observations is $\bar{x}$, if first term is increased by 1 second term is increased by 2 and so on. What will be the new mean?



We apply the respective formulae depending upon the information provided in the problem.

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